COPY #1

GETTING STARTED

COPY can be run by any FLEX user without reading this manual. How can such a claim be possible? COPY is compatible with the command syntax of the copy program supplied with FLEX, thus if you can us FLEX you can use this program. You are urged to assemble the source program provided and start using COPY even before read past this page of the manual. Though you will be using only a part of COPY capability you will have many hidden advantages such as error checking, alphabetizing, and data retention. When you have time to read the remainder of this manual you will learn how to use the many options supported by COPY.

ASSEMBLING THE SOURCE

COPY is provided as a source file that can be assembled by the FLEX assembler for 6800 and 6809 processors. The only change that need be made for CPU is one character in the line:

```
FLEX EQU  $A000
```

Locate this line using your favorite editor and confirm that it matches the FLEX you are using. The FLEX starting address should be $A000 for 6800 or $C000 for 6809. Note only one character need be changed. There are optional changes that will be explained later. COPY can now be assembled and the object should be named COPY.CMD. You may want to retain your old COPY program under a different name until you gain confidence in the new program.

INITIAL SURPRISES

COPY will accept commands in the same format as the original copy, but you will immediately notice differences in how the program performs. There will be changes in timing, such as pauses when sorting the directory and half as many head loads per file copied. You will notice files are copied in alphabetical order. The original creation date of the file being copied will be used for the destination file. If your disk has bad files you may get some unfamiliar error messages. COPY has a lot of error checking which will uncover problems you may not be aware of. The most common error is:

```
BAD DATE: FILENAME.EXT    SIZE   DATE
ALLOW BAD DATES  (Y/N)?
```

Throughout this manual these terms will be used often:

COPY	the new program
OLD COPY	the program provided with FLEX
SOURCE	the file being copied from (any type of file)
DESTINATION	the file being written to
OPTION LETTERS	command line letters used to enable options
MATCH LIST	list of starting characters (see below)

it will be assumed that you are familiar with the syntax of the copy command provided with FLEX. All FLEX users know that the old copy program supports a match string list for copying multiple files that match the letter in the list. The command:

```
COPY 0 1 AB X .CM
```

will copy from drive zero to drive one all files that start with "AB" or "X" or have the extensions that start with "CM". This is the most common syntax you will use with COPY.
Many FLEX user do not know that the old copy program supports a useful syntax for copying a single file. "P.CMD" without including other commands files starting with "P" you will find a match list copy unsuitable. The command:

```
COPY 1.P.CMD 2
```

Will copy only one file from drive one to drive two. A variation os this single copy file copy is used to copy one file and change the name:

```
COPY 0.FLEX2.SYS 1.FLEXNEW.SYS
```

In entering any COPY command either a comma or a space can be used to separate parameters.

OPTION LETTERS

There are over a dozen option letters supported in COPY which makes it very powerful and a bit overwhelming. It is important to know that you do not need to understand all of the options you may not want to use. If you can't remember the letter for the option you need, you can type COPY without any parameters and the program will display a HELP list of option letters with a short explanation of each. Option letters are included on the command line after the word COPY and before the drive number or file name:

```
COPY LDN 0,1 A.TXT
```

Any number of letter may be used in any order. If an unimplemented letter is used COPY will display the HELP list of options. Options are positive; that is using the letter will enable the option. This contrasts with the confusion of FLEX assembler which some negative options. You may recall that assembler options +NS means YES include line numbers but NO do not include symbol table. Much of the confusion with the this otherwise excellent assembler is due to the ambiguous method of handling option letters. It was decided to avoid ambiguity by having all option letters enable the desired option. If a letter is accidently used twice the option is still enabled.

DEFAULT OPTIONS

COPY has default options which take effect whenever there are no option letters on the command line. The most common default option will be "A" which enables sorting of the source directory before files are copied. COPY has provisions for changing the default options in the assembler source program for those users who want other options defaulted. For example one could default only option "U" and COPY will emulate the old copy by using the current FLEX date on the destination file. If the user must change the default options there is a table near the start of COPY which contains the default option bytes. Each byte is FCB 0 if the option is not defaulted and FCB 1 if it is defaulted. If the user needs to verify which options are defaulted, typing COPY without parameters will display an example followed by all the options. The defaulted options will be marked with "YES":

```
Example: COPY ADNL 0,1 .CMD A

YES  A  copy in Alphabetical order
NO   C  allow Corrupt files to be copied
NO   D  will copy files with newer date
NO   E  delete Existing destination file
NO   F  copy by file number (alpha not allowed)
NO   K  Kill duplicate file on source
NO   L  List files without copying
NO   N  copy files Not on destination
NO   O  turn Off defaults
NO   P  Prompt before copying file
NO   R  Recover from track-sector
NO   S  makes Second copy of file (.CPY ext)
NO   U  Use current FLEX date
NO   W  Wait for disk change
NO   Z  Zap source file after copying
```

EXPLANATION OF OPTION LETTERS

A -COPY IN ALPHABETICAL ORDER

The "A" option enables a sort subroutine that will alphabetize the source directory before files are selected to be copied. Most users will have this as a default option so the files will be in order when cataloged. If you enter any option on the command line all defaults are turned off. If option "A" was defaulted and you want to alphabetize you must add "A" to the options you enter.

C -ALLOW CORRUPT FILES TO BE COPIED

The "C" option will enable you to copy a file that is damaged by a CRC error or record sequence error. This is a slightly dangerous option which should only be used if you don't have an alternate copy of a file.

D -COPY FILES WITH A NEWER DATE

The "D" option will find files that are on both disks and compare their creation dates. If the source file is newer it will be copied as a replacement for the older destination file.

E -DELETE EXISTING DESTINATION DATE

The "E" option is used when you want to replace a file on the destination disk. This option will suppress the prompt for whether you wish to delete the existing file. It will often be used along with "D" to update a disk with newer versions of programs.

F -COPY BY FILE NUMBER

The "F" option changes copy's parameters from a match string list to a list of file numbers. Follow the drive numbers with a list of file numbers for those files that you want to copy. File numbers can be found with a utility such as DIR.CMD (or CAT +M command). A group of files can be specified as a starting and ending number separated by a dash. The command "COPY F 0,1 5 13-18 9" will copy file five, files thirteen through eighteen, and file nine.

K -KILL DUPLICATE FILE ON SOURCE

The "K" option is very dangerous. This command isn't really a copy, rather it uses the directory compare routines to delete files from source disk that appear on the destination. This allows you to clear off extra copies of programs not needed on the source disk. It operates very fast and will clear a number of files faster than you can hit reset. As with all dangerous options it is protected with an "ARE YOU SURE" prompt. "COPY K 1,0" is most effective in killing files on drive 1 when they exist on drive 0. "COPY KD 1,0" will kill the file on drive 1 when it is older than the file on drive 0. Use COPY KDL 1,0 to preview what files will be deleted.

L -LIST WITHOUT COPYING

The "L" option disables the file copy subroutine. This is used to display a list of files that WOULD have been copied if you hadn't used option "L". This can be used with other options to check disk for duplicate files, newer dates, bad files etc.

N -COPY FILES NOT ON DESTINATION

The "N" option is used to copy the files on the source disk that are not already on the destination disk. This can be used to add all new files to a backup disk.

O -TURN OFF DEFAULT OPTIONS

The "O" option is a dummy character used to turn off all default options if you don't want any options. If used with any other letters it has no effect.

P prompt BEFORE COPYING

The "P" option enables this prompt: "Prompt off (P):FLEX (F):copy (Y/N)?". you should respond with "P" if you want to continue copying without the prompt or "F" to return to FLEX or "Y" to copy this file. "N" or any other character will skip to the next file. This is useful for scanning trough a disk copying only certain files. Another use is skipping down to a certain file on a disk and copying all files after that.

R -RECOVER FROM TRACK-SECTOR

The "R" option is used to read a file without using the directory. If the directory of a disk has been destroyed but the user knows the files starting address, the file can be recovered. The command "COPY R 1 2B 50.newfile" will read from drive 1, track $2B, sector 5 until encountering an end of file or a record out of sequence. The write file extension will default to .SCR. A second use for this option is to recover a deleted file. The first sector of a deleted file can be found with the new CAT.CMD. If the file has not been over-written, COPY can recover it. Record sequence checking eliminates the restriction that the file be the last file deleted. Provision was made to start copying in the middle of a file. This is a somewhat dangerous option since it allows the user to override the FLEX file manager. The command must be typed as shown with three numbers and a file name. No other options should be used with "R".

S -MAKE Second COPY OF A FILE

The "S" option is used when you want two copies of a file to be written. The second copy will have the sane name as the first with the extension ".CPY". This is useful when sending a program to a friend whose drive may have difficulty reading your disk. By sending two copies there is a much higher chance the file will be readable.

U -USE CURRENT FLEX DATE

The "U" option is used when you want the destination file to have the present FLEX date rather than the date of the source file being copied. This may be useful if you know the source file has an error date.

W -WAIT FOR DISK CHANGE

The "W" option eliminates the need to copy COPY.CMD onto the source disk so that you may remove your system disk to insert the destination disk. When "W" is used on the command line COPY will wait for a key to be pressed before accessing any directories.

Z -ZAP SOURCE FILE AFTER COPYING

The "Z" option is somewhat dangerous. It is used to delete the file from the source disk after it is copied. Essentially the file moved from one disk to the other.

EXAMPLES

Often options can be combined to perform tasks that previously were impossible or required a separate command utility. Some examples:

COPY C 0.BAD.TXT 1 Copies one file that has a CRC error, including whatever data is readable from the bad sector. Replaces the utility RAWCOPY.CMD.

COPY DN 0,1 Updates the destination with all files from the source that are not on the destination or have an older date on the destination. Replaces the utilities COPYNEW and ARCHIVE.

COPY EZ 0.P.CMD 2 Moves file P.CMD from source to destination no matter what. Replaces the utility MV.CMD.

COPY F 2,0 1-16 Useful for copying from one large COPY F 2,1 17-52 drive to smaller ones. Copies 16 files to drive zero and the remainder to drive one.

COPY KD 1,0 Kill the file on drive 1 when it is older than the file on drive 0.

COPY LNA 2,1 Alphabetically lists those files on the source that are not on the destination. Replaces the utility DUP.CMD and is much faster.

COPY P 0,1 Prompt before each file is copied to allow user to select which files are desired. Replaces the utility PCOPY.

COPY R 2 4 6 1.f00 Recover a file on drive 2 that does not appear in the directory. Reading will begin at track 4 sector 6 and the new file will be written to drive 1 with the name F00.SCR. Replaces the utility RECOVER.cmd.

COPY W 0,1 .CMD After COPY is loaded the user is prompted with "Change disk-press key". The system disk may be removed and another disk inserted.

DANGEROUS OPTIONS

A few options letters enables dangerous functions such as Killing the original file on the source drive. These options should only be used by an "Experienced Copier" and are protected with this prompt:
dangerous option selected are you sure (Y/N)?
If you accidently stumble on a dangerous option type "N" and you will be returned to FLEX. With proper use these options can be quite useful.

ERROR DETECTION

COPY has many error checks that are not found in other FLEX utilities. Many FLEX users have found that disk problems often surface when using old copy, often with disastrous results. One error COPY traps is files with a size of zero sectors. These files usually result from pressing reset while doing file operations. Novice FLEX users do not realize that it they reboot FLEX after aborting with reset a defective file on the disk. When a program like TSC EDIT.CMD opens a file, FLEX creates a directory entry with zero sectors. If the program is aborted with reset or the disk is removed the zero sector file remains in the directory. Later, when the disk is copied, the defective file will result in entry free space of the disk being copied. This copy "runaway" results in the destination disk being clobbered. Another problem which is eliminated with COPY results from bad link in the source file causing it to intersect with another file. A third error solved by COPY is a bad link on the destination disk causing the file to overwrite the directory. A time wasting error in the old copy was it would try to copy a file that wouldn't fit in the available space on the disk. Copy will catch this error and report "file will no fit".
Copy has several of the common FLEX error messages built in. These were included since many users of two drive systems will need to remove their system disk to copy files. This results in FLEX reporting errors by number since the file ERRORS.SYS will not be found on the system drive. Also convenience the drive associated with the error is reported along with the offending track and sector. This is useful for such errors as CRC which could occur on either the source or destination drive, and the file or the directory.
The most common error message reported by users of COPY is "DATE BAD". This occurs when the user does not enter a valid date when FLEX is booted or by failure of a hardware clock when used for setting the FLEX date. Copy will check the dates on the files when it reads the disk directory and report any dates outside a reasonable range. This reduces the chance that a bad date will be passed on the new file. There are two alternatives when the "BAD DATE" appears. You can answer "Y" indicating that you approve of bad dates or answer "N" and not copy the file. After returning to FLEX you can re-enter COPY using option "U" which will assign the current FLEX date to the file or use the new date command to set the file date to the day the file was made.

NON STANDARD FLEX ROUTINES

Considerable care was made to determine that COPY would work on all FLEX systems. All accesses to the disk were made through the File Manager System to avoid compatibility problems with user written disk drivers. Still it was necessary to use a few tricks in order to make the program more useful. The FMS does not provide a means for determining if a disk is protected or even if it exists without attempting to write on it. This is the FIRST thing a program should do so the user doesn't waste time answering prompts only find the program abort later. COPY reads the SIR (sector 3) of the destination disk then duplicates it on sector 4. This is an unused sector on all systems tested. If the disk is protected or not ready the program will exit immediately. COPY shuts off interrupts when accessing track zero to avoid problems with the spooler. COPY checks the next to the last byte ($FE) of the SIR for extended directory flag. This byte is set non-zero by some users to indicate the catalog has been extended to include all of track 1. If this flag is set COPY will protect track one as well as track zero. Finally the file date is inserted into the destination file FCB before the file is closed. This is done to keep the original date on the new file.

PRINTING AND SPOOLING

Occasionally one may want to keep a log of the files being copied perhaps for unattended operation. The command "P.COPY 0,1" will print the file specifications as the files are copied. A duplicate of each line appear on the terminal along with prompts. The command "0 OUTFILE COPY 0,1" will do the same for a disk file. Since the OUTFILE will be open when COPY reads the directory do not be alarmed that it is reported as a BAD FILE.

COMMENT ON DATES AND FILE NAMES

Many FLEX users are lax in entering the correct date when booting their system. Part of the responsibility for this lies in the old copy program which used the current FLEX date when copying a file, thus rendering the file date almost useless. With this new COPY utility the date takes a new significance since the date reflects when the file was originally created.